Coefficient of Association

Yule's Coefficient of Association measures the strength and direction of association. "Association" means that the attributes have some degree of agreement.

2×2 Contingency Table

Attribute A ⇩	Attribute B		Total
	Yes B	No β	
Yes A	(AB)	$(A\beta)$	(A)
No α	(αB)	$(\alpha\beta)$	(α)
Total	(B)	(β)	N

Yule's coefficient: $Q = \dfrac{(AB)(\alpha\beta) - (A\beta)(\alpha B)}{(AB)(\alpha\beta) + (A\beta)(\alpha B)}$

Note 1: The usage of the symbol α is not to be confused with level of significance.

Note 2: (AB): Number with attributes AB etc.

This coefficient ranges from –1 to +1. The values between –1 and 0 indicate inverse relationship (association) between the attributes. The values between 0 and +1 indicate direct relationship (association) between the attributes.

Example 4.7

Out of 1800 candidates appeared for a competitive examination 625 were successful; 300 had attended a coaching class and of these 180 came out successful. Test for the association of attributes attending the coaching class and success in the examination.

Solution:

$N = 1800$

A: Success in examination

α: No success in examination

B: Attended the coaching class

β: Not attended the coaching class

$(A) = 625,\ (B) = 300,\ (AB) = 180$

	B	β	Total
A	180	445	625
α	120	1055	1175
Total	300	1500	$N = 1800$

12th_Statistics_EM_Unit_4.indd 121 1/28/2020 6:34:33 PM